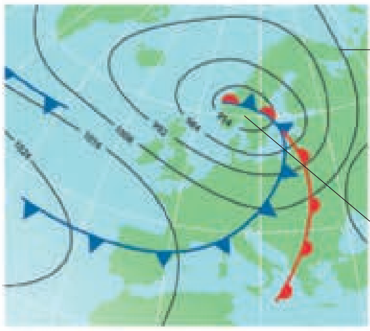
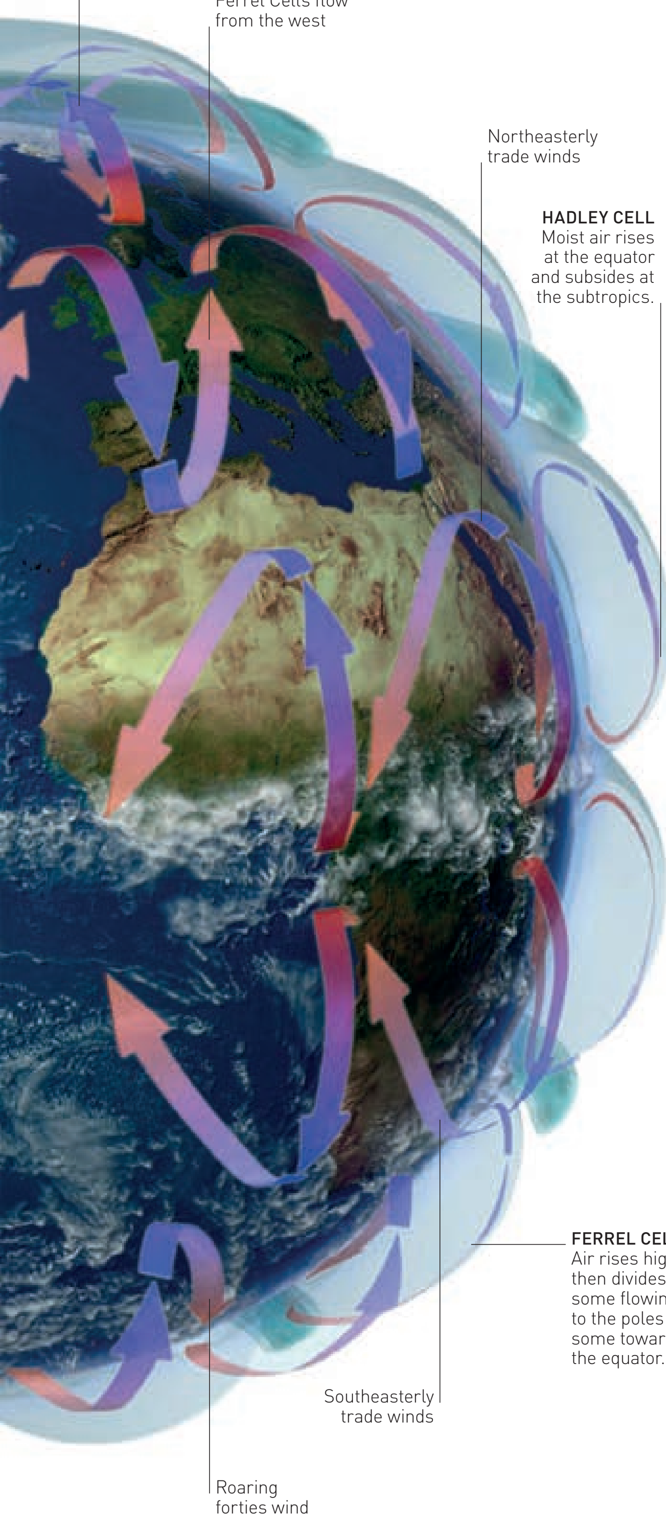
STATIONARY FRONT
This occurs where air masses push against one another without moving. Can cause heavy rain.

OCCLUDED FRONT
Fast-moving cold air overtakes a warm front, and lifts the warm air mass. It can rain for days.



A WEATHER MAP

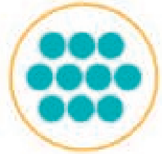


PRECIPITATION

All precipitation is simply falling moisture. Whether water falls from a cloud as rain, hail, or snow depends on how cold the air is.

SNOW
Snowflakes are clusters of frozen water droplets.

RAIN
Droplets in clouds fall when they are too heavy to float.



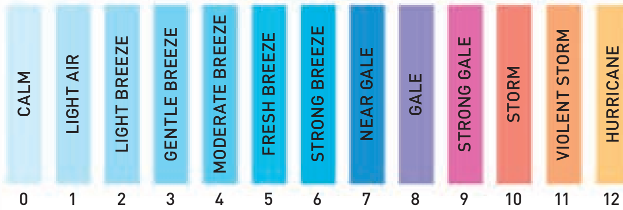
HAIL
Ice pellets form from crystals in storm clouds.

SLEET
Sleet is a mix of snow and rain in air above freezing.

FOG
Fog forms when warm, moist air hits cold ground.

WIND

Air moving between high and low pressure areas is called wind. Wind speed – from still air to a hurricane – is measured on the Beaufort scale.



BEAUFORT SCALE

RECORD-BREAKING WEATHER

Some places have extreme climates, or weather events that are talked about for years.

WINDIEST
The fastest wind speed in a tornado was 450 km/h (280 mph), recorded at Wichita Falls, USA, in 1958.

HOTTEST
The hottest land-surface temperature ever recorded (by satellite measurement) was 70.7°C (159.3°F) in the Lut Desert, Iran, in 2005.

COLDEST
The coldest recorded temperature was –93°C (–136°F), measured in Antarctica's eastern highlands in 2010.

WETTEST
The highest rainfall recorded in one day was 18.25 cm (71.9 in) in Foc-Foc, Reunion Island, in the Indian Ocean in 1966, during a tropical cyclone.

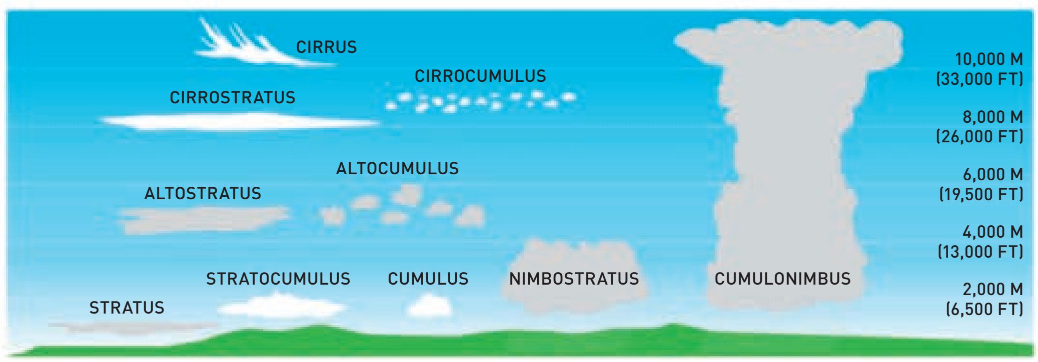
DRIEST
Arica, Chile is the populated area with the lowest average annual rainfall in the world at 0.76 mm (0.03 in).

ARICA, CHILE

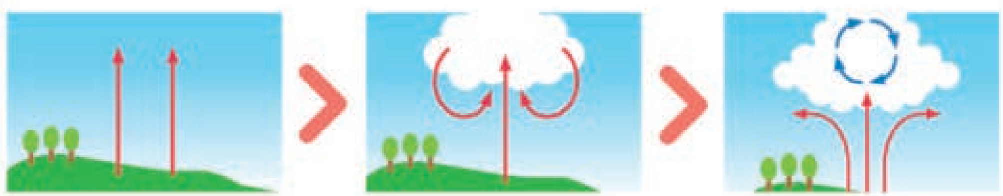
CLOUDS

All clouds fall into three main groups, although each type has many different shapes. Cumulus form pillowy heaps; stratus have flat layers; and cirrus are wispy streaks.

CLOUD NAMES
Clouds are named according to their size, shape, and height.



HOW CLOUDS FORM



1 **MOISTURE** When the Sun shines, moisture in the ground and sea rises into the air as water vapour.

2 **VAPOUR CONDENSES** As the vapour rises and cools, it condenses and forms clouds made of tiny water droplets.

3 **CLOUDS RISE** During cloud formation, heat is released into the surrounding air and lifts the cloud higher.

WATCHING THE WEATHER

Weather stations are at work all over the world, gathering information about local and global weather patterns. They use a range of instruments from simple thermometers and rain gauges to weather balloons and satellites, which use sensors to monitor Earth's atmosphere.

DOPPLER RADAR
This type of radar uses microwaves to track moving bands of rain.

WEATHER BALLOON
Helium-filled balloons carry sensors high into the atmosphere.

SATELLITE
Satellites orbit from pole to pole or sit above one region.